

第十章 非线性力学初步

线性:

$$\vec{F} = m\vec{r} \rightarrow F \text{ 为基本变量的线性函数}$$

$L(q, \dot{q})$ 最多为二次项

§ 10.1 非线性振动与微扰法

$$V(x) = V_0 + \frac{1}{2} kx^2 + \frac{1}{6} \alpha x^3 + \frac{1}{24} \beta x^4$$

$$T(x) = \frac{1}{2} m \dot{x}^2$$

$$L = T - V = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} kx^2 - \frac{1}{6} \alpha x^3 - \frac{1}{24} \beta x^4 - V_0$$

拉格朗日方程: $\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = 0$

$$\hookrightarrow m\ddot{x} + kx + \alpha x^2 + \beta x^3 = 0$$

0) 零级: $m\ddot{x}_0 = -kx_0$ ↑ - 阶 + ↑ - 阶 +

$$\hookrightarrow x_0 = A \cos(\sqrt{\frac{k}{m}}t + \varphi)$$

1) 一级: $x = x_0 + x_1$ x_1 为 - 阶小

$$m\ddot{x}_0 + m\ddot{x}_1 = -k(x_0 + x_1) - \alpha(x_0^2 + 2x_0x_1 + x_1^2) - \beta(x_0^3 + 3x_0^2x_1 + 3x_0x_1^2 + x_1^3)$$

$$\hookrightarrow m\ddot{x}_1 = -kx_1 - \alpha x_0^2$$
 含 $\frac{1}{2}$ = 阶小

$$= -kx_1 - \frac{1}{2} \alpha A^2 \cos^2(\omega_0 t + \varphi)$$

$$= -kx_1 - \frac{1}{4} \alpha A^2 \{1 + \cos 2(\omega_0 t + \varphi)\}$$

$$\hookrightarrow \ddot{x}_1 + \omega_0^2 x_1 + \frac{1}{2m} \alpha A^2 + \frac{1}{2m} \alpha A^2 \cos 2(\omega_0 t + \varphi) = 0$$

设 $x_1 = A_1 \cos(\omega_1 t + \varphi_1) + B_1$

$$-\omega_1^2 A_1 \cos(\omega_1 t + \varphi_1) + \omega_0^2 A_1 \cos(\omega_1 t + \varphi_1) + \omega_0^2 B_1 + \frac{1}{2m} \alpha A^2 + \frac{1}{2m} \alpha A^2 \cos 2(\omega_0 t + \varphi) = 0$$

$$\{(\omega_0^2 - \omega_1^2)\} A_1 \cos(\omega_1 t + \varphi_1) + \frac{1}{2m} \alpha A^2 \cos 2(\omega_0 t + \varphi) + \omega_0^2 B_1 + \frac{1}{2m} \alpha A^2 = 0$$

$$\hookrightarrow \begin{cases} \omega_1 = 2\omega_0, \varphi_1 = 2\varphi, A_1 = \frac{\frac{1}{2m} \alpha A^2}{\omega_0^2 - \omega_1^2} = \frac{\alpha A^2}{6m\omega_0^2} \\ B_1 = -\frac{\alpha A^2}{2m\omega_0^2} \end{cases}$$

$$x_1 = \frac{\alpha A^2}{6m\omega_0^2} \cos 2(\omega_0 t + \varphi) - \frac{\alpha A^2}{2m\omega_0^2} = \frac{\alpha A^2}{6m\omega_0^2} \{ \cos 2(\omega_0 t + \varphi) - 2 \}$$

$$x = A \cos(\omega_0 t + \varphi) + \frac{\alpha A^2}{6m\omega_0^2} \cos 2(\omega_0 t + \varphi) - \frac{\alpha A^2}{2m\omega_0^2}$$

设 $x = x_0 + x_1 + x_2$ x_2 为二阶小量

$$m\ddot{x}_0 + m\ddot{x}_1 + m\ddot{x}_2 = -kx_0 - kx_1 - kx_2 - \alpha(x_0^2 + x_1^2 + x_2^2 + 2x_0x_1 + 2x_0x_2 + 2x_1x_2) \\ - \beta(x_0^3 + x_1^3 + x_2^3 + 3x_0^2x_1 + 3x_0x_1^2 + 3x_0^2x_2 + 3x_0x_2^2 + 3x_1^2x_2 + 3x_1x_2^2 + 6x_0x_1x_2)$$

$$\Rightarrow m\ddot{x}_2 = -kx_2 - 2\alpha x_0x_1 - \beta x_0^3$$

$$= -kx_2 - 2\alpha A \cos(\omega_0 t + \varphi_0) \left[\cos 2(\omega_0 t + \varphi_0) - 1 \right] - \beta A^3 \cos^3(\omega_0 t + \varphi_0)$$

$$\cos x \cos 2x = \frac{1}{2}(\cos 3x + \cos x)$$

$$\cos^3 x = \frac{1}{2}(1 + \cos 2x) \cos x = \frac{3}{4} \cos x + \frac{1}{4} \cos 3x$$

$$= -kx_2 - \frac{A^3 \alpha^2}{3m\omega_0^2} \cos 2(\omega_0 t + \varphi_0) \left[\cos 2(\omega_0 t + \varphi_0) - 1 \right] - \beta A^3 \cos^3(\omega_0 t + \varphi_0)$$

$$= -kx_2 + \left(\frac{3}{6} \frac{A^3 \alpha^2}{m\omega_0^2} - \frac{3}{4} \beta A^3 \right) \cos(\omega_0 t + \varphi_0)$$

$$= \left(\frac{A^3 \alpha^2}{6m\omega_0^2} + \frac{1}{4} \beta A^3 \right) \cos 3(\omega_0 t + \varphi_0)$$

$$\Rightarrow \ddot{x}_2 + \omega_0^2 x_2 = \left(\frac{3A^3 \alpha^2}{6m\omega_0^2} - \frac{3\beta A^3}{4m} \right) \cos(\omega_0 t + \varphi_0) + \left(\frac{A^3 \alpha^2}{6m\omega_0^2} + \frac{\beta A^3}{4m} \right) \cos 3(\omega_0 t + \varphi_0)$$

$$x_2 = A_2 \cos(\omega_2 t + \varphi_2) + B_2$$

$$(\omega_2^2 - \omega_0^2) A_2 \cos(\omega_2 t + \varphi_2) + \omega_0^2 B_2 = \left(\frac{3A^3 \alpha^2}{6m\omega_0^2} - \frac{3\beta A^3}{4m} \right) \cos(\omega_0 t + \varphi_0)$$

$$+ \left(\frac{A^3 \alpha^2}{6m\omega_0^2} + \frac{\beta A^3}{4m} \right) \cos 3(\omega_0 t + \varphi_0) = 0$$

$\Rightarrow A_2 = \infty$? 共振不再为 ω_0 有一小变化: $\omega = \omega_0 + \omega_2$

$$= \overline{\omega} + \omega_2$$

$$x_0 = A \cos(\omega t + \varphi_0)$$

$$x_1 = -\frac{A^2 \alpha}{2m\omega^2} + \frac{A^2 \alpha}{6m\omega^2} \cos 2(\omega t + \varphi_0)$$

$$x = A \cos(\omega t + \varphi_0) - \frac{A^2 \alpha}{2m\omega^2} + \frac{A^2 \alpha}{6m\omega^2} \cos 2(\omega t + \varphi_0) + x_2$$

$$\ddot{x}_2 + \omega^2 x_2 = \left(\frac{3A^3 \alpha^2}{6m\omega^2} - \frac{3\beta A^3}{4m} \right) \cos(\omega t + \varphi_0) + \left(\frac{A^3 \alpha^2}{6m\omega^2} + \frac{\beta A^3}{4m} \right) \cos 3(\omega t + \varphi_0)$$

$$x = A \cos(\omega t + \varphi_0) - \frac{A^2 \alpha}{2m\omega^2} + \frac{A^2 \alpha}{6m\omega^2} \cos 2(\omega t + \varphi_0) + x_2$$

2mω₀ mω₀²

$$m\ddot{x} = -kx - \alpha x^2 - \beta x^3$$

$$\begin{aligned} & -m\omega^2 A \cos(\omega t + \varphi_0) - \frac{2A^2 d^2}{2m\omega^2} \cos(2\omega t + \varphi_0) + m\ddot{x}_0 \\ & + k[A \cos(\omega t + \varphi_0) - \frac{A^2 d}{2m\omega^2} + \frac{A^2 d}{6m\omega^2} \cos(2\omega t + \varphi_0) + x_2] \\ & + \alpha[A \cos(\omega t + \varphi_0) - \frac{A^2 d}{2m\omega^2} + \frac{A^2 d}{6m\omega^2} \cos(2\omega t + \varphi_0) + x_2]^2 \\ & + \beta[A \cos(\omega t + \varphi_0) - \frac{A^2 d}{2m\omega^2} + \frac{A^2 d}{6m\omega^2} \cos(2\omega t + \varphi_0) + x_2]^3 = 0 \end{aligned}$$

保留至 1 阶 +

$$\begin{aligned} \hookrightarrow & -m\omega^2 A \cos(\omega t + \varphi_0) - 2m\omega_0 \omega_r A \cos(\omega t + \varphi_0) \\ & - \frac{2A^2 d^2}{2} \cos(2\omega t + \varphi_0) + m\ddot{x}_0 \end{aligned}$$

$$m\ddot{x}_0 = -kx_0$$

$$m\ddot{x}_1 = -kx_1 - \alpha x_1^2$$

$$\begin{aligned} & + k[A \cos(\omega t + \varphi_0) - \frac{A^2 d}{2m\omega^2} + \frac{A^2 d}{6m\omega^2} \cos(2\omega t + \varphi_0) + x_2] \\ & + \alpha[A \cos(\omega t + \varphi_0) - \frac{A^2 d}{2m\omega^2} + \frac{A^2 d}{6m\omega^2} \cos(2\omega t + \varphi_0)]^2 \\ & + \frac{A^3 d^2}{3m\omega^2} \cos(\omega t + \varphi_0) [\cos(2\omega t + \varphi_0) - 3] + \beta A^3 \cos^3(\omega t + \varphi_0) \end{aligned}$$

→

$$\begin{aligned} & m\ddot{x}_1 + kx_1 - (2m\omega_0 \omega_r A + \frac{A^3 d^2}{m\omega^2}) \cos(\omega t + \varphi_0) \quad \cos \alpha \cos 2\alpha = \frac{1}{2} \cos \alpha + \frac{1}{2} \cos 3\alpha \\ & + \frac{A^3 d^2}{6m\omega^2} \cos(\omega t + \varphi_0) + \frac{A^3 d^2}{6m\omega^2} \cos 3(\omega t + \varphi_0) \quad \cos^3 \alpha = \frac{3}{4} \cos \alpha + \frac{1}{4} \cos 3\alpha \\ & + \frac{3}{4} \beta A^3 \cos(\omega t + \varphi_0) + \frac{1}{4} \beta A^3 \cos 3(\omega t + \varphi_0) \end{aligned}$$

ω ≠ ω₀ 也是共振项?

$$= m\ddot{x}_1 + kx_1 + (\frac{3}{4} \beta A^3 - \frac{5A^3 d^2}{6m\omega^2} - 2m\omega_0 \omega_r A) \cos(\omega t + \varphi_0)$$

$$+ (\frac{1}{4} \beta A^3 + \frac{A^3 d^2}{6m\omega^2}) \cos 3(\omega t + \varphi_0) = 0 \quad \text{共振项} = 0$$

$$\Rightarrow \frac{3}{4} \beta A^3 - \frac{5A^3 d^2}{6m\omega^2} - 2m\omega_0 \omega_r = 0$$

$$\text{解得 } \omega_r = \frac{A^2 (9\beta m\omega^2 - 10d^2)}{24m^2\omega^2}$$

§ 10.2 参数共振

简谐振动参数共振时, $m = m(t)$, $k = k(t)$

$$\frac{d}{dt}(m\dot{x}) + kx = 0$$

$$\frac{1}{m(t)} \frac{d}{dt} \left[m(t) \cdot \left(\frac{1}{m(t)} \frac{d}{dt} x \right) \right] + k(t)x = 0 \quad \frac{dt}{dt} = \frac{d\tau}{m(t)}, \quad \frac{d}{dt} = \frac{1}{m(t)} \frac{d}{d\tau}$$

$$\hookrightarrow \frac{d^2 x}{d\tau^2} + \underbrace{m(t)k(t)}_{f(\tau)} x = 0 \quad f(\tau) \text{ 为周期函数} \rightarrow f(t) \text{ 为周期函数?}$$

讨论(1) 如 $\ddot{x} + \omega^2(t)x = 0$ 的方程

其中 $\omega(t)$ 为一周期函数, 可近似为 $\omega^2(t) = \omega_0^2 (1 + q \cos \Omega t)$

$$\ddot{x} + \omega_0^2 (1 + q \cos \Omega t) x = 0$$

$$\text{设 } \Omega = 2\omega_0 + \Delta\omega \quad - \text{ 附注} \quad \Omega_1 = 2\omega_0 + \Delta\omega, \quad \Omega_2 = \frac{2\omega_0}{2} + \Delta\omega, \quad \Omega_3 = \frac{2\omega_0}{3} + \Delta\omega$$

猜解(1) 为 $x = A(t) \cos(\omega_0 + \frac{1}{2}\Delta\omega)t + B(t) \sin(\omega_0 + \frac{1}{2}\Delta\omega)t$

$$\hookrightarrow -(\omega_0 + \frac{1}{2}\Delta\omega)^2 A(t) \cos(\omega_0 + \frac{1}{2}\Delta\omega)t - (\omega_0 + \frac{1}{2}\Delta\omega)^2 B(t) \sin(\omega_0 + \frac{1}{2}\Delta\omega)t$$

$$- 2(\omega_0 + \frac{1}{2}\Delta\omega) \dot{A}(t) \cos(\omega_0 + \frac{1}{2}\Delta\omega)t + 2(\omega_0 + \frac{1}{2}\Delta\omega) \dot{B}(t) \sin(\omega_0 + \frac{1}{2}\Delta\omega)t$$

$$+ \ddot{A}(t) \cos(\omega_0 + \frac{1}{2}\Delta\omega)t + \ddot{B}(t) \sin(\omega_0 + \frac{1}{2}\Delta\omega)t$$

$$+ \omega_0^2 [1 + q \cos(2\omega_0 + \Delta\omega)t] [A(t) \cos(\omega_0 + \frac{1}{2}\Delta\omega)t + B(t) \sin(\omega_0 + \frac{1}{2}\Delta\omega)t] = 0$$

$$\hookrightarrow -(\omega_0^2 + \omega_0 \Delta\omega + \frac{1}{4} \Delta\omega^2) A(t) \cos(\omega_0 + \frac{1}{2}\Delta\omega)t$$

$$- (\omega_0^2 + \omega_0 \Delta\omega + \frac{1}{4} \Delta\omega^2) B(t) \sin(\omega_0 + \frac{1}{2}\Delta\omega)t$$

$$- (\omega_0 + \frac{1}{2}\Delta\omega) \dot{B}(t) \sin(\omega_0 + \frac{1}{2}\Delta\omega)t + (\omega_0 + \frac{1}{2}\Delta\omega) \dot{A}(t) \cos(\omega_0 + \frac{1}{2}\Delta\omega)t$$

$$+ \ddot{A}(t) \cos(\omega_0 + \frac{1}{2}\Delta\omega)t + \ddot{B}(t) \sin(\omega_0 + \frac{1}{2}\Delta\omega)t$$

$$+ \omega_0^2 A(t) \cos(\omega_0 + \frac{1}{2}\Delta\omega)t + \omega_0^2 B(t) \sin(\omega_0 + \frac{1}{2}\Delta\omega)t$$

$$+ \frac{1}{2} \omega_0^2 q A \cos(\omega_0 + \frac{1}{2}\Delta\omega)t + \frac{1}{2} \omega_0^2 q A \cos 3(\omega_0 + \frac{1}{2}\Delta\omega)t$$

$$+ \frac{1}{2} \omega_0^2 q B \sin 3(\omega_0 + \frac{1}{2}\Delta\omega)t - \frac{1}{2} \omega_0^2 q B \sin(\omega_0 + \frac{1}{2}\Delta\omega)t$$

$$\hookrightarrow \cos(\omega_0 + \frac{1}{2}\Delta\omega)t \left[-A(t) (\omega_0^2 + \omega_0 \Delta\omega + \frac{1}{4} \Delta\omega^2) + (2\omega_0 + \Delta\omega) \dot{B}(t) \right.$$

$$\left. + \ddot{A}(t) + \omega_0^2 A(t) + \frac{1}{2} \omega_0^2 q A \right] + \sin 3(\omega_0 + \frac{1}{2}\Delta\omega)t \left[\frac{1}{2} \omega_0^2 q B \right.$$

$$+ \ddot{B}(t) + \omega_0^2 B(t) - \frac{1}{2} \omega_0 q B(t) \Big]_{=0} + \sin 3(\omega_0 + \frac{1}{2} \Delta \omega) t \frac{1}{2} \omega_0 q B = 0$$

略去高频项和二阶小, $\ddot{A}, \ddot{B}, \dot{A} \Delta \omega, \dot{B} \Delta \omega \leftarrow$ 求导一次小一阶?

$$\begin{cases} 2\omega_0 \dot{B}(t) + \omega_0 (\frac{1}{4} \omega_0 q - \Delta \omega) A(t) = 0 \\ -2\omega_0 \dot{A}(t) - \omega_0 (\frac{1}{2} \omega_0 q + \Delta \omega) B(t) = 0 \end{cases}$$

$$\eta = \begin{bmatrix} A \\ B \end{bmatrix} \quad K = \begin{bmatrix} 0 & -\frac{1}{4} \omega_0 q - \frac{\Delta \omega}{2} \\ \frac{\Delta \omega}{2} - \frac{1}{4} \omega_0 q & 0 \end{bmatrix}$$

$$\dot{\eta} = K \eta$$

$$\text{取 } \eta = \eta_0 e^{i\omega t}$$

$$i\omega \eta_0 e^{i\omega t} = K \eta_0 e^{i\omega t} \rightarrow \omega \text{ 为 } K \text{ 的特征值}$$

$$\hookrightarrow [-i\omega I + K] \eta = 0$$

$$\text{为使 } \eta \text{ 有非平凡解: } |-i\omega I + K| = \begin{vmatrix} -i\omega & -\frac{1}{4} \omega_0 q - \frac{i\Delta \omega}{2} \\ \frac{i\Delta \omega}{2} - \frac{1}{4} \omega_0 q & -i\omega \end{vmatrix} = 0$$

$$\text{解得 } \omega = \left[\left(\frac{1}{4} \omega_0 q \right)^2 - \left(\frac{1}{2} \Delta \omega \right)^2 \right]^{1/2}$$

$$\text{参数共振: } \omega \text{ 为实数} \rightarrow |\Delta \omega| < \frac{1}{2} \omega_0 q$$

$$\hookrightarrow \Omega \in \left(\left(2 - \frac{1}{2} q \right) \omega_0, \left(2 + \frac{1}{2} q \right) \omega_0 \right)$$

考虑有阻力情况:

$$\eta = \eta_0 e^{(i\omega - \beta)t}$$

$$\text{振幅增大: } \omega > \beta \rightarrow |\Delta \omega| < \left[(2\beta)^2 - \left(\frac{1}{2} \omega_0 q \right)^2 \right]^{1/2}$$

$$\hookrightarrow \Omega \in \left(2\omega_0 - \left[(2\beta)^2 - \left(\frac{1}{2} \omega_0 q \right)^2 \right]^{1/2}, 2\omega_0 + \left[(2\beta)^2 - \left(\frac{1}{2} \omega_0 q \right)^2 \right]^{1/2} \right)$$

$$\text{且 } q < \frac{4\beta}{\omega_0}$$

$$\omega(t+T) = \omega(t)$$

$$\hookrightarrow x_1(t+T) = \mu_1 x_1(t), \quad x_2(t+T) = \mu_2 x_2(t)$$

$$\text{解得 } x_1 = \mu_1^{t/T} \gamma_1(t), \quad x_2 = \mu_2^{t/T} \gamma_2(t)$$

其中 $\gamma_1(t), \gamma_2(t)$ 为时间的周期函数

$$\begin{cases} \ddot{x}_1 + \omega(t)x_1 = 0 \\ \ddot{x}_2 + \omega(t)x_2 = 0 \end{cases} \rightarrow \dot{x}_1 x_2 - \dot{x}_2 x_1 = \frac{d}{dt} (x_2 x_1 - x_1 x_2) = 0$$

$$\begin{aligned} & \dot{x}_2(t+T)x_1(t+T) - x_1(t+T)\dot{x}_2(t+T) \\ &= \mu_1 \mu_2 (\dot{x}_2(t)x_1(t) - x_1(t)\dot{x}_2(t)) \\ &= \dot{x}_2(t)x_1(t) - x_1(t)\dot{x}_2(t) \end{aligned}$$

$$\Rightarrow \mu_1 \mu_2 = 1$$

$$\ddot{x} + \underline{\omega(t)} x = 0 \quad \omega \in \mathbb{R}$$

$$\hookrightarrow \ddot{x}^* + \omega(t)x^* = 0 \quad \mu_1, \mu_2 \Leftrightarrow \mu_1^*, \mu_2^*$$

$$\textcircled{1} (\mu_1, \mu_2) = (\mu_1^*, \mu_2^*)$$

$$\text{令 } \mu = \mu_1, \mu_2 = \frac{1}{\mu}$$

$$\Rightarrow x_1 = \mu^{t/T} \gamma_1(t), \quad x_2 = \mu^{-t/T} \gamma_2(t)$$

$$(i) |\mu| < 1, \quad x_1 \rightarrow \infty \quad (t \rightarrow \infty)$$

$$(ii) |\mu| < 1, \quad x_2 \rightarrow \infty \quad (t \rightarrow \infty)$$

$$\textcircled{2} (\mu_1, \mu_2) = (\mu_2^*, \mu_1^*)$$

$$\mu_1 \mu_2 = \mu_1 \mu_1^* = |\mu_1|^2 = 1 \quad \therefore |\mu_1| = |\mu_2| = 1$$

§ 10.3 平衡点、极限环和轨道稳定性

相空间 $\xrightarrow{\text{平衡点附近}}$ 相平面.

1.

右平衡点, $\frac{\partial H}{\partial p_2} = \dot{q}_2 = 0, -\frac{\partial H}{\partial q_2} = \dot{p}_2 = 0$

设 $\dot{q} = f(q, p), \dot{p} = g(q, p)$ 不含时: 自治系统

泰勒展开: $f = \frac{\partial f}{\partial q} \bigg|_0 q + \frac{\partial f}{\partial p} \bigg|_0 p$

$$\dot{q} = a_{11} q + a_{12} p, \quad \dot{p} = a_{21} q + a_{22} p$$

试探解: $\eta = \begin{bmatrix} q \\ p \end{bmatrix} = \begin{bmatrix} q_0 e^{\lambda t} \\ p_0 e^{\lambda t} \end{bmatrix} = \eta_0 e^{\lambda t}$

$$\dot{\eta} = A \eta$$

$$\hookrightarrow \lambda \eta = A \eta, \quad (\lambda I - A) \eta = 0$$

$$\text{非零解: } \begin{vmatrix} \lambda - a_{11} & -a_{12} \\ -a_{21} & \lambda - a_{22} \end{vmatrix} = \lambda^2 - (a_{11} + a_{22}) \lambda + (a_{11} a_{22} - a_{12} a_{21}) = 0$$

解得 $\lambda_1 = \quad, \lambda_2 = \quad$

$$\eta = \eta_{10} e^{\lambda_1 t} + \eta_{20} e^{\lambda_2 t}$$

$$\begin{cases} q = c_1 A_1 e^{\lambda_1 t} + c_2 A_2 e^{\lambda_2 t} \\ p = c_1 B_1 e^{\lambda_1 t} + c_2 B_2 e^{\lambda_2 t} \end{cases}$$

① λ_1, λ_2 为同号实根

$\lambda_1, \lambda_2 > 0$ 远离平衡点

$\lambda_1, \lambda_2 < 0$ 靠近平衡点

② $\lambda_1 = \lambda_2^* \rightarrow \eta_{10} = \eta_{20}^*$

$$\text{所设 } \lambda_1 = \lambda_a + i \lambda_b, \quad \lambda_2 = \lambda_a - i \lambda_b$$